

 	Republic of the Philippines Department of Transportation
	Document No: <u>DOTr-HRDD-Forms-004</u> Rev. No.: <u>001</u> Effective Date: <u>03 August 2022</u>
	RE-ENTRY ACTION PLAN (REAP) FORM

INSTRUCTIONS: This form is used in preparing a proposed Re-entry Action Plan in applying to a Scholarship Program. Duly accomplished and signed forms will be submitted to the HRD Division along with the other documentary requirements.

Title of Scholarship Program:	Japanese Grant Aid for Human Resource Development Scholarship (JDS)
Name of the Development Partner	Japanese International Cooperation Center (JICE)

I. REAP BACKGROUND AND INFORMATION	
a. REAP Title	Social Signals: Analyzing Railway Service Quality in the Philippines Using Social Media Data and Natural Language Processing (NLP)
b. REAP Objective	The study aims to improve the quality of railway transportation service by analyzing commuter complaints and feedback on key service attributes such as but not limited to punctuality, cleanliness, safety, and customer service. This will be achieved by using Natural Language Processing (NLP) to process user-generated data from social media platforms such as X (formerly Twitter) and Facebook within 2 years. By extracting actionable insights from public feedback and commuter sentiments shared online, the methodology will focus on leveraging public engagement and the social representation of commuters to address specific service shortcomings. The derived insights will be used to formulate or amend existing public policies to make them more passenger centric.
c. Problem/Opportunity	<i>What issue or opportunity does your REAP hope to address in the Host Organization?</i> <i>Cite concrete examples, cases, and/or reports to support your view that this important but not urgent problem or opportunity exists.</i> Railway systems are vital for promoting efficient, effective, and sustainable transportation in densely populated regions. In the Philippines, specifically in Metro Manila, railways such as MRT-3 serve as a lifeline for millions of commuters daily. Currently, various improvements are being implemented such as rehabilitation, capacity building for technical staff, as well as the introduction of new technologies. However, these operational metrics cannot fully leverage passenger satisfaction, as there are social dimensions that need to be addressed such as inclusivity, accessibility, equity, safety and security. Addressing these challenges requires concrete tools for assessing railway service quality.

	Traditional data collection methods, such as customer satisfaction surveys and public interviews, are constrained by limited representation, functional objectives, costs, and delayed feedback mechanisms, leading to a gap in timely actionable insights (Nguyen <i>et al.</i>, 2022). On the other hand, social media platforms like X (formerly Twitter) and Facebook became outlets for passengers to share their experience, feedback, and frustrations. These platforms offer real-time and unsolicited information about passenger sentiment and experiences, making them a rich yet untapped source of data. Currently, the ROs only utilize social media by interacting with users, such as responding to comments and posts; however, the actual actions being taken remain unclear to users. Previous studies, such as (Zhijing Jin and Rada Mihalcea 2023), have highlighted the potential of social media data in assessing public perceptions and improving service delivery. Despite the potential of social media for understanding public perceptions, systematic approaches to leveraging this data for railway service quality assessment are still lacking in the Philippine context. Natural Language Processing (NLP) provides an effective means of addressing this gap. NLP, a branch of Data Science, specializes in quantitative analysis of unstructured text data to extract meaningful patterns, trends, and sentiment. By applying NLP to social media data, we can identify emerging issues at scale (Nguyen <i>et al.</i>, 2022). As demonstrated by (Wang and Taylor, 2016), NLP applications in Architecture, Engineering, and Construction (AEC) research can be used in uncovering critical insights from social media. This approach can provide valuable insights into passenger expectations, helping railway operators and policymakers prioritize improvements and enhance service quality. Despite its potential, research on applying NLP to analyze railway service quality remains limited, particularly in a developing country like the Philippines. Previous studies in transportation have primarily focused on structured data, such as surveys, ticketing records, or information systems, leaving a gap in exploring passenger-generated content on social media. This research aims to address these gaps by leveraging social media data to analyze the service quality of the Philippine railway system using NLP. It will explore passenger sentiments, recurring and emerging issues, and best practices on key social dimensions such as safety, accessibility, and reliability, providing a data-driven perspective on their experiences. Additionally, it seeks to identify actionable insights that can inform improvements in railway operations and contribute to public policy development. By integrating modern analytical methods with social media data, this study aspires to complement traditional service quality assessments and foster a more responsive and passenger-centric approach to railway management. Moreover, this research holds the potential to revolutionize how railway service quality and policies can be improved in the Philippines, making the system more aligned with commuter needs while contributing to global studies on transportation and analytics.
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d. Description in relation to the Department i. Organizational Outcome	<i>Determine at least one REAP outcome. How will your REAP help the Host Organization enhance service delivery, policies, systems, or competencies?</i> The Philippine Railways Institute is the training and research arm of the DOTr. One of its mandates is to leverage the human resources of the railway sector. Based on the REAP discussed earlier, one potential outcome is the institutionalization of a data-driven framework for evaluating and improving railway service quality. This framework will enable PRI to develop quality interventions to address the needs and gaps of the railway sector, ensure the railway personnel competence, and develop quality research for the continuous improvement of the railway workforce to maintain an efficient, reliable, and safe railway transportation services. <i>Please indicate which of the following organizational outcomes will your REAP address:</i> 1) Policy Evaluation and Formulation REAP will expand the feedback mechanism used by PRI in formulating and evaluating policies. By utilizing social media and NLP, it will create opportunities for more data-driven policy research that is more aligned to the people. These efforts will also create avenues in exploring digital transformation initiatives such as investment to the current information systems of PRI to fully institutionalize these advancements. Additionally, since we are establishing partnerships across different sectors, such as railway service providers, academe, training institutes, and government agencies in developing policies and training, this research will be a part of our portfolio, showcasing our commitment in achieving our goals. 2) Enhancement of Service Delivery The REAP will contribute to the policy research and development which is anchored to PRI's training and development interventions of the railway personnel. The insights that will be derived such as emerging and untapped social issues of passengers in the railway system will influence the improvement of transportation policies and service delivery. This chain of effect will create a holistic pipeline to continuously improve transportation services based on the real-time feedback of commuters.
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ii. Target Competency Improvement	<i>Which competencies do you expect to gain or improve as a result of your studies? How will your suggested program of study contribute to the development of these competencies? You must articulate the significance of your proposed program of study in providing the competencies required to execute your REAP in a clear and succinct manner.</i> 1. Public Policy Analysis Understanding policy context will support me in understanding the environment of policymaking and service delivery within railway transportation. This knowledge will ensure that the findings in my research are aligned in the railway sector and not just for academics. Thus, gaining this competency is pivotal, because it will bridge data insights and policy implementation. 2. Social Science Since this study focuses on leveraging public engagement and social representation, improving my knowledge in social science is important. By analyzing tweets from X and Facebook comments, I need to delve into human behavior, social dynamics, equity issues that affect the satisfaction of service delivery in railway transportation. Additionally, to anchor with public policy analysis, I need to bridge these two competencies to ensure that I know what implications of my study in the society. 3. Application of Data Science to Policy Development By undertaking this study and enrolling in the Asian Public Policy Program at Hitotsubashi University, it is imperative to improve this competency. Data Science, in which NLP was part of, is a primary tool in various domains such as advertising and marketing is use. However, since it involves social media and personal preferences, I need to establish a limitation (especially in data collection) so that only practical applications are explored, ensuring there is no violations of ethical standards particularly in data privacy.
e. Relation to the Government's Development Plan	<i>The REAP targeting the problem or opportunity of the Host Organization is aligned with the goals set in the Philippine Government's Development Plan</i> The objective of the REAP is to improve the railway service quality through the utility of public engagement and social representation in social media. The application of NLP in this data, leverage the traditional methodology in order to deliver passenger-oriented recommendations, responsive and timely action, and data-driven results. This is anchored under "Building A High Trust Society" of the Philippine Development Plan (PDP) to ensure responsiveness, people centered, technology-enabled, and clean governance. Strategies like proactive sourcing of citizen's perspective in order to broaden participatory governance.

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f. Direct Beneficiary	<i>Who will see immediate advantages from your REAP's implementation? How many target beneficiaries does your REAP have? How will your REAP benefit your intended beneficiaries?</i> The direct beneficiaries of this REAP are the commuters and the human resources of the railway sector. Since commuters generate the data for the analysis, future improvements in railway services are expected to align with commuter preferences. This is anticipated to enhance confidence in and patronage of railway services. As the data will be utilized by PRI, it will contribute to policymaking and serve as a basis for training enhancements benefiting railway personnel. Furthermore, with the expected increase in patronage, workforce expansion is likely, thereby strengthening the human resources of the railway sector.
g. Gender Equality, Disability, and Social Inclusion (GEDSI)	<i>How will the implementation of your REAP help women and other marginalized groups (e.g., IP, PWD, LGBTQ++, etc.)? Provide specific strategies to address the needs of the various groups.</i> The utility of social media data provides its users with an accessible space to share their experiences, voice their opinions, and highlight specific issues they encounter when using railway transportation. The utilization of user-generated data leverages public engagement and diversity, creating opportunities to generate passenger-oriented recommendations. This is expected to support the adoption of a more inclusive approach to transportation planning, ensuring that women and marginalized groups feel safer, more valued, and better served in their daily commutes.
h. Output	<i>What output(s) will your REAP generate? How will these outcomes help achieve the desired organizational outcome?</i> 1. Passenger-oriented Policy Research By utilizing user-generated data, policy research and trainings will be improved. This will create a more effective curriculum for the capacity development of the human resources of the railway sector. 2. Railway Service Performance Dashboard Using the data collection system of the REAP, a performance dashboard can be developed to monitor real-time feedback. Thus, assisting the ROs in timely action in addressing passenger concerns.

	This innovative intervention will support efficient, reliable, and safe railway transportation service.	
II. REAP TIMELINE AND IMPLEMENTATION		
a. Measurability	<i>How will you monitor the success of your REAP implementation? What are your milestones?</i> 1. Commuter Sentiment. Improvement of feedback by comparing the railway service sentiments from each data collection and analysis. 2. Ridership Diversity. Improvement of diversity of ridership of the railway lines. It can be monitored in the same manner as the methodology since demographics are present in data collection. 3. Operational Efficiency. Reduction in the average time taken to respond to passenger-reported issues to the ROs. 4. Reduction in Complaint Recurrence. Percentage decrease in repeat complaints about the same issues after implementing corrective measures.	
b. Sustainability	<i>What is your plan for ensuring that the REAP's gains are institutionalized and sustained after the program's completion?</i> 1. Establishment of an information system that will operationalize the REAP. This will ensure reproducibility and fosters knowledge sharing. 2. Capacity Development of PRI staff in incorporating data-driven research in policy formulation and evaluation. 3. Establishment of reliable data collection such as good feedback system. 4. Embedding the REAP methodology to PRIs research and development. 5. Establishment of partnerships with private firms and academes to pursue application of data science and machine learning to the human resources of railway sector.	
c. Predictive Action Steps with Identified Success Indicators		
i. Phase/Stage	ii. Expected Output	iii. Timeline

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Data Collection	1. Social Media Dataset – 2 years' worth of data composed of posts, comments, or feedback about service quality of the three railway lines (MRT3, LRT1, LRT2). 2. Documented and reusable Python scripts for data extraction, with annotations explaining the methodology. <i>Tools: web scraper, twitter/facebook API</i>	3 weeks to 1 month (21 – 30 days)
Data Processing	1. Railway related tweets/comments with words reduced to roots words, uncapitalized, filtered from special characters, links, profanity, and politically inclined idea. <i>Tools: Pandas, Natural Language Toolkit (NLTK), Excel</i>	3 weeks to 1 month (21 – 30 days)
Data Analysis and Modeling	1. Quantitative sentiment scores (positive or negative) for posts about railway services. 2. Topic Modeling Results – key topics or issues from social media discussions, generated using methods like Non-negative Matrix Factorization, Latent Dirichlet Allocation, or BERTopic.	3 weeks to 1 month (21 – 30 days)

	3. Key Performance Indicators (KPIs) 4. Classification models 5. Evaluation Metrics Tools: ScikitLearn, Power BI	
Results Discussion	1. Data Visualization (word clouds, sentiment graphs, correlation maps). 2. Interpretation of results - explanation of identified sentiments and topics in relation to railway service quality. 3. Policy Recommendations - Passenger centric suggestions for improving railway services. 4. Final Research Paper	3 weeks to 1 month (21 – 30 days)
d. Target Date to Start	November 2026	
e. Target Date to End	March 2027	
III. RESOURCES		
Estimate the budget you'll need (in pesos per budget item) to effectively implement your REAP. Include your plan for obtaining the required financial assistance. 1. API subscription fee for X and Facebook – Php 3000 (3 months) 2. Storage and Processing: Amazon Web Service (AWS) Subscription - \$ 29 (monthly) or \$ 299 (annual) 3. Training and Capacity Building – Php 10,000		

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IV. COMMENT(S) AND RECOMMENDATION(S)	
a. Immediate Superior (Mentor)	-
b. Head of the Office/Division/Service	
c. Human Resource Development Division	

Prepared by:



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PATRICK JOSE C. ROXAS

Name of Applicant with signature

Recommended by:

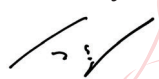


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EISEN LOWELL A. GOCO

Immediate Supervisor with signature (Mentor)

Approved and noted by:



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USEC. JEREMY S. REGINO

OIC - Undersecretary and Executive Director for

Philippines Railways Institute